

IGNOU BCS-012: Basic Mathematics
Term-End Examination, December 2024
Full Solved Question Paper with Final Theory Sheet
 Time: 3 Hours Maximum Marks: 100

Solved Paper

Question 1(a)

Show that

$$\begin{vmatrix} 1 & a & a^2 \\ a^2 & 1 & a \\ a & a^2 & 1 \end{vmatrix} = (a^3 - 1)^2.$$

Solution. Expanding along the first row,

$$\begin{aligned} D &= 1 \begin{vmatrix} a & a^2 \\ a^2 & 1 \end{vmatrix} - a \begin{vmatrix} a^2 & a \\ a & 1 \end{vmatrix} + a^2 \begin{vmatrix} a^2 & 1 \\ a & a^2 \end{vmatrix} \\ &= (1 - a^3) - a(a^2 - a^2) + a^2(a^4 - a) \\ &= 1 - a^3 + a^6 - a^3 = a^6 - 2a^3 + 1 = (a^3 - 1)^2. \end{aligned}$$

Hence proved.

Question 1(b)

If

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix},$$

show that $A^2 - 4A - 5I_3 = O_3$.

Solution. First compute

$$A^2 = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix}.$$

Also,

$$4A + 5I_3 = \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} = A^2.$$

Therefore $A^2 - 4A - 5I_3 = O_3$.

Question 1(c)

Use the principle of mathematical induction to show that $3^{2n} - 1$ is divisible by 8 for each natural number n .

Solution. Let $P(n)$ be the statement that $3^{2n} - 1$ is divisible by 8.

For $n = 1$,

$$3^2 - 1 = 9 - 1 = 8,$$

which is divisible by 8. Thus $P(1)$ is true.

Assume $P(k)$ is true, i.e., $3^{2k} - 1 = 8m$ for some integer m . Then

$$\begin{aligned} 3^{2(k+1)} - 1 &= 9 \cdot 3^{2k} - 1 \\ &= 9(3^{2k} - 1) + 8. \end{aligned}$$

By the induction assumption, $3^{2k} - 1$ is divisible by 8, so $9(3^{2k} - 1) + 8$ is also divisible by 8. Hence $P(k+1)$ is true. Therefore, by induction, $3^{2n} - 1$ is divisible by 8 for all $n \in \mathbb{N}$.

Question 1(d)

If in an A.P. $a = 2$ and the sum of the first five terms is one fourth of the sum of next five terms, show that $a_{20} = -112$.

Solution. Let the common difference be d . The first term is $a = 2$.

$$S_5 = \frac{5}{2}[2a + 4d] = \frac{5}{2}[4 + 4d] = 10 + 10d.$$

The sum of the next five terms is $a_6 + a_7 + a_8 + a_9 + a_{10} = S_{10} - S_5$.

$$S_{10} = \frac{10}{2}[2a + 9d] = 5[4 + 9d] = 20 + 45d.$$

Thus,

$$S_{10} - S_5 = (20 + 45d) - (10 + 10d) = 10 + 35d.$$

Given $S_5 = \frac{1}{4}(S_{10} - S_5)$,

$$10 + 10d = \frac{1}{4}(10 + 35d).$$

So,

$$40 + 40d = 10 + 35d \Rightarrow 5d = -30 \Rightarrow d = -6.$$

Therefore,

$$a_{20} = a + 19d = 2 + 19(-6) = 2 - 114 = -112.$$

Hence proved.

Question 1(e)

If $z \in \mathbb{C}$, $|z| = 1$, $z \neq -1$, show that

$$\frac{z-1}{z+1}$$

is purely imaginary.

Solution. Since $|z| = 1$, we have $z\bar{z} = 1$, hence $\bar{z} = \frac{1}{z}$. Let

$$w = \frac{z-1}{z+1}.$$

Then

$$\bar{w} = \frac{\bar{z}-1}{\bar{z}+1} = \frac{\frac{1}{z}-1}{\frac{1}{z}+1} = \frac{1-z}{1+z} = -\frac{z-1}{z+1} = -w.$$

If $\bar{w} = -w$, then $w + \bar{w} = 0$, so $\operatorname{Re}(w) = 0$. Hence w is purely imaginary.

Question 1(f)

If two roots r_1 and r_2 of $x^2 + kx + 12 = 0$ are such that $(r_1 - r_2) = 1$, find k .

Solution. For $x^2 + kx + 12 = 0$,

$$r_1 + r_2 = -k, \quad r_1 r_2 = 12.$$

Given $r_1 - r_2 = 1$. Squaring,

$$(r_1 - r_2)^2 = (r_1 + r_2)^2 - 4r_1 r_2.$$

Therefore,

$$1 = (-k)^2 - 4(12) = k^2 - 48.$$

Hence

$$k^2 = 49 \Rightarrow k = \pm 7.$$

So the possible values are

$$\boxed{k = 7 \text{ or } k = -7.}$$

Question 1(g)

Two tailors A and B earn Rs. 600 per day and Rs. 800 per day respectively. Tailor A can stitch 6 shirts and 4 pants per day, while Tailor B can stitch 10 shirts and 4 pants per day. How many days shall each work if it is desired to produce at least 60 shirts and 32 pants at minimum labour cost? Also calculate the least cost.

Solution. Let x be the number of days Tailor A works and y be the number of days Tailor B works. We minimize

$$C = 600x + 800y$$

subject to

$$6x + 10y \geq 60, \quad 4x + 4y \geq 32, \quad x \geq 0, \quad y \geq 0.$$

Equivalently,

$$3x + 5y \geq 30, \quad x + y \geq 8.$$

The corner points of the feasible region relevant for minimization are:

$$(10, 0), \quad (0, 8), \quad \text{and intersection of } 3x + 5y = 30, \quad x + y = 8.$$

Solving the intersection:

$$x + y = 8 \Rightarrow x = 8 - y.$$

Substitute in $3x + 5y = 30$:

$$3(8 - y) + 5y = 30 \Rightarrow 24 + 2y = 30 \Rightarrow y = 3,$$

so $x = 5$.

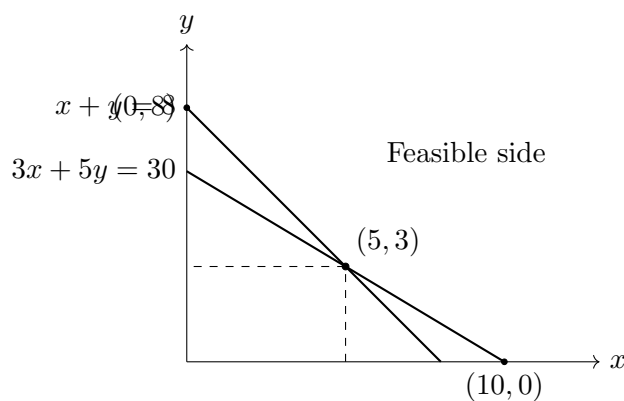
Cost at the corner points:

$$C(10, 0) = 6000, \quad C(0, 8) = 6400, \quad C(5, 3) = 600(5) + 800(3) = 5400.$$

Thus the minimum cost is obtained at $(x, y) = (5, 3)$.

Tailor A should work 5 days and Tailor B should work 3 days.

Minimum cost = Rs. 5400.

**Question 2(a)**

Using elementary row operations, find inverse of

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}.$$

Solution. Using row operations on $[A \mid I]$:

$$\left[\begin{array}{ccc|ccc} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 2 & 3 & 0 & 1 & 0 \\ 3 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & -4 & 3 & -1 \\ 0 & 0 & 1 & \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{array} \right].$$

Therefore,

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix}.$$

Question 2(b)

How many terms of the G.P. $\sqrt{3}, 3, 3\sqrt{3}, \dots$ add up to $39 + 13\sqrt{3}$?

Solution. Here $a = \sqrt{3}$ and $r = \sqrt{3}$. The sum of n terms is

$$S_n = \frac{a(r^n - 1)}{r - 1} = \frac{\sqrt{3}((\sqrt{3})^n - 1)}{\sqrt{3} - 1}.$$

Checking the first six terms:

$$\sqrt{3} + 3 + 3\sqrt{3} + 9 + 9\sqrt{3} + 27 = 39 + 13\sqrt{3}.$$

Hence,

$$\boxed{n = 6.}$$

Question 2(c)

If $1, \omega, \omega^2$ are cube roots of unity, show that

$$(2 + 3\omega + 2\omega^2)^9 (4 + 3\omega + 3\omega^2)^9 = 1.$$

Solution. For cube roots of unity,

$$1 + \omega + \omega^2 = 0, \quad \omega^3 = 1.$$

Thus $\omega^2 = -1 - \omega$. Now,

$$2 + 3\omega + 2\omega^2 = 2 + 3\omega + 2(-1 - \omega) = \omega,$$

$$4 + 3\omega + 3\omega^2 = 4 + 3\omega + 3(-1 - \omega) = 1.$$

Therefore,

$$(2 + 3\omega + 2\omega^2)^9 (4 + 3\omega + 3\omega^2)^9 = \omega^9 \cdot 1^9 = (\omega^3)^3 = 1.$$

Hence proved.

Question 2(d)

If α and β are roots of $3x^2 - 4x + 1 = 0$, form an equation whose roots are $\frac{\alpha^2}{\beta}$ and $\frac{\beta^2}{\alpha}$.

Solution. For $3x^2 - 4x + 1 = 0$,

$$\alpha + \beta = \frac{4}{3}, \quad \alpha\beta = \frac{1}{3}.$$

Let the new roots be

$$p = \frac{\alpha^2}{\beta}, \quad q = \frac{\beta^2}{\alpha}.$$

Their product is

$$pq = \frac{\alpha^2\beta^2}{\alpha\beta} = \alpha\beta = \frac{1}{3}.$$

Their sum is

$$p + q = \frac{\alpha^3 + \beta^3}{\alpha\beta}.$$

Now,

$$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = \left(\frac{4}{3}\right)^3 - 3\left(\frac{1}{3}\right)\left(\frac{4}{3}\right) = \frac{64}{27} - \frac{4}{3} = \frac{28}{27}.$$

Thus,

$$p + q = \frac{28/27}{1/3} = \frac{28}{9}.$$

Hence the required equation is

$$x^2 - \frac{28}{9}x + \frac{1}{3} = 0,$$

or

$$\boxed{9x^2 - 28x + 3 = 0}.$$

Question 3(a)

Find

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+3} - \sqrt{3}}{x}.$$

Solution. Rationalize:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{x+3} - \sqrt{3}}{x} &= \lim_{x \rightarrow 0} \frac{(\sqrt{x+3} - \sqrt{3})(\sqrt{x+3} + \sqrt{3})}{x(\sqrt{x+3} + \sqrt{3})} \\ &= \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{x+3} + \sqrt{3})} \\ &= \frac{1}{2\sqrt{3}}. \end{aligned}$$

Therefore,

$$\boxed{\frac{1}{2\sqrt{3}}}.$$

Question 3(b)

If $y = ae^{mx} + be^{-mx}$, show that

$$\frac{d^2y}{dx^2} = m^2y.$$

Solution. Differentiate once:

$$\frac{dy}{dx} = ame^{mx} - bme^{-mx}.$$

Differentiate again:

$$\frac{d^2y}{dx^2} = am^2e^{mx} + bm^2e^{-mx} = m^2(ae^{mx} + be^{-mx}) = m^2y.$$

Hence proved.

Question 3(c)

Determine the interval in which

$$f(x) = -2x^3 + 24x + 7$$

is increasing.

Solution. A function is increasing where $f'(x) > 0$. Here,

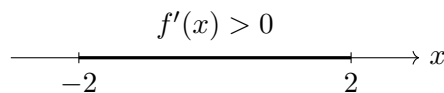
$$f'(x) = -6x^2 + 24 = 6(4 - x^2).$$

So

$$f'(x) > 0 \iff 4 - x^2 > 0 \iff x^2 < 4 \iff -2 < x < 2.$$

Therefore, $f(x)$ is increasing on

$$\boxed{(-2, 2)}.$$

**Question 3(d)**

If a mothball evaporates at a rate proportional to its surface area $4\pi r^2$, show that its radius decreases at a constant rate.

Solution. The volume of a spherical mothball is

$$V = \frac{4}{3}\pi r^3.$$

Since evaporation rate is proportional to surface area and volume decreases,

$$\frac{dV}{dt} = -k(4\pi r^2), \quad k > 0.$$

Differentiate V with respect to t :

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}.$$

Therefore,

$$4\pi r^2 \frac{dr}{dt} = -k(4\pi r^2).$$

For $r \neq 0$,

$$\frac{dr}{dt} = -k.$$

Thus the radius decreases at the constant rate k .

Question 4(a)

Evaluate

$$\int \frac{x}{\sqrt{x+3}} dx.$$

Solution. Put $u = x + 3$, so $x = u - 3$ and $du = dx$. Then

$$\begin{aligned} \int \frac{x}{\sqrt{x+3}} dx &= \int \frac{u-3}{\sqrt{u}} du \\ &= \int (u^{1/2} - 3u^{-1/2}) du \\ &= \frac{2}{3}u^{3/2} - 6u^{1/2} + C. \end{aligned}$$

Substituting $u = x + 3$,

$$\int \frac{x}{\sqrt{x+3}} dx = \frac{2}{3}(x-6)\sqrt{x+3} + C.$$

Question 4(b)

Evaluate

$$\int_a^b \frac{\log x}{x} dx, \quad a > b > 0.$$

Solution. Let $u = \log x$. Then $du = \frac{1}{x}dx$. Hence

$$\int_a^b \frac{\log x}{x} dx = \int_{\log a}^{\log b} u du = \frac{1}{2} [(\log b)^2 - (\log a)^2].$$

Therefore,

$$\int_a^b \frac{\log x}{x} dx = \frac{1}{2} [(\log b)^2 - (\log a)^2].$$

Question 4(c)

Find the area bounded by $y = \sqrt{x}$ and $y = x$.

Solution. The curves intersect where

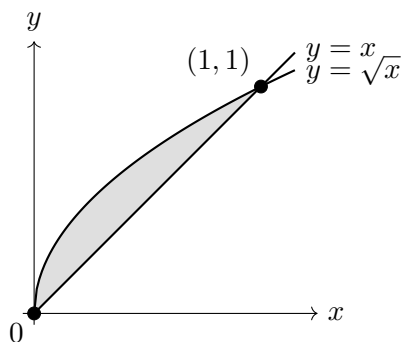
$$\sqrt{x} = x.$$

Since $x \geq 0$, squaring gives $x = x^2$, so $x = 0$ or $x = 1$. On $[0, 1]$, $\sqrt{x} \geq x$. Therefore, the required area is

$$A = \int_0^1 (\sqrt{x} - x) dx = \left[\frac{2}{3}x^{3/2} - \frac{x^2}{2} \right]_0^1 = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}.$$

Thus,

$$A = \frac{1}{6} \text{ square units.}$$

**Question 4(d)**

If $\vec{a}$ and $\vec{b}$ are two vectors, show that

$$|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|.$$

Solution. This is the triangle inequality. We prove it using dot product:

$$\begin{aligned} |\vec{a} + \vec{b}|^2 &= (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) \\ &= |\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2. \end{aligned}$$

Since $\vec{a} \cdot \vec{b} \leq |\vec{a}| |\vec{b}|$,

$$|\vec{a} + \vec{b}|^2 \leq |\vec{a}|^2 + 2|\vec{a}| |\vec{b}| + |\vec{b}|^2 = (|\vec{a}| + |\vec{b}|)^2.$$

Taking square roots,

$$\boxed{|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|.}$$

Question 5(a)

Show that the lines

$$\frac{x-5}{4} = \frac{y-7}{4} = \frac{z+3}{-5}$$

and

$$\frac{x-8}{7} = \frac{y-4}{1} = \frac{z-5}{3}$$

intersect.

Solution. Write the first line as

$$x = 5 + 4t, \quad y = 7 + 4t, \quad z = -3 - 5t.$$

Write the second line as

$$x = 8 + 7s, \quad y = 4 + s, \quad z = 5 + 3s.$$

For intersection, solve

$$5 + 4t = 8 + 7s, \quad 7 + 4t = 4 + s, \quad -3 - 5t = 5 + 3s.$$

The solution is

$$t = -1, \quad s = -1.$$

Substituting in the first line,

$$x = 5 + 4(-1) = 1, \quad y = 7 + 4(-1) = 3, \quad z = -3 - 5(-1) = 2.$$

Substituting in the second line gives the same point:

$$x = 8 + 7(-1) = 1, \quad y = 4 + (-1) = 3, \quad z = 5 + 3(-1) = 2.$$

Thus the lines intersect at

$$\boxed{(1, 3, 2).}$$

Question 5(b)

Solve the inequality

$$\left| \frac{x-3}{2} \right| \leq 1.$$

Solution.

$$\left| \frac{x-3}{2} \right| \leq 1 \iff -1 \leq \frac{x-3}{2} \leq 1.$$

Multiplying by 2,

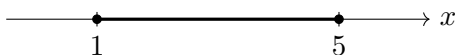
$$-2 \leq x-3 \leq 2.$$

Adding 3,

$$1 \leq x \leq 5.$$

Therefore,

$$x \in [1, 5].$$

**Question 5(c)**Using determinants, find the area of a triangle whose vertices are $(1, 2)$, $(-2, 3)$ and $(-3, -4)$.**Solution.** The area of a triangle with vertices (x_1, y_1) , (x_2, y_2) and (x_3, y_3) is

$$\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}.$$

Thus,

$$\begin{aligned} \Delta &= \frac{1}{2} \begin{vmatrix} 1 & 2 & 1 \\ -2 & 3 & 1 \\ -3 & -4 & 1 \end{vmatrix} \\ &= \frac{1}{2} |1(3+4) - 2(-2+3) + 1(8+9)| \\ &= \frac{1}{2} |7 - 2 + 17| = \frac{1}{2} (22) = 11. \end{aligned}$$

Therefore,

$$\text{Area} = 11 \text{ square units.}$$

Question 5(d)

If

$$y = \log_e \left[e^x \left(\frac{x-2}{x+2} \right)^{3/4} \right],$$

find $\frac{dy}{dx}$.**Solution.** Using logarithmic properties,

$$y = \log_e(e^x) + \log_e \left[\left(\frac{x-2}{x+2} \right)^{3/4} \right] = x + \frac{3}{4} \log_e \left(\frac{x-2}{x+2} \right).$$

So,

$$\begin{aligned}\frac{dy}{dx} &= 1 + \frac{3}{4} \left(\frac{1}{x-2} - \frac{1}{x+2} \right) \\ &= 1 + \frac{3}{4} \left(\frac{4}{x^2-4} \right) \\ &= 1 + \frac{3}{x^2-4}.\end{aligned}$$

Therefore,

$$\boxed{\frac{dy}{dx} = 1 + \frac{3}{x^2-4}.$$

Theory and Formula Sheet for This Paper

Important theory, formulas and methods

Determinants

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

For a 3×3 determinant, expand along a convenient row or column. Row/column operations may simplify the determinant.

Matrix inverse To find A^{-1} using elementary row operations:

$$[A \mid I] \rightarrow [I \mid A^{-1}].$$

If $A^2 - 4A - 5I = O$, then $A^2 = 4A + 5I$.

Mathematical induction To prove $P(n)$ for all $n \in \mathbb{N}$:

1. Prove $P(1)$.
2. Assume $P(k)$.
3. Prove $P(k+1)$.

A.P.

$$a_n = a + (n-1)d, \quad S_n = \frac{n}{2}[2a + (n-1)d].$$

G.P.

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad (r \neq 1).$$

Quadratic roots For $ax^2 + bx + c = 0$:

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}.$$

Also,

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta.$$

Complex numbers If $|z| = 1$, then

$$z\bar{z} = 1, \quad \bar{z} = \frac{1}{z}.$$

A number w is purely imaginary if

$$\bar{w} = -w.$$

Cube roots of unity

$$1 + \omega + \omega^2 = 0, \quad \omega^3 = 1.$$

Limits Rationalize expressions containing square roots:

$$\frac{\sqrt{x+a} - \sqrt{a}}{x} = \frac{1}{\sqrt{x+a} + \sqrt{a}}.$$

Differentiation

$$\frac{d}{dx}e^{mx} = me^{mx}, \quad \frac{d}{dx}e^{-mx} = -me^{-mx}.$$

A function is increasing where $f'(x) > 0$.

Sphere

$$V = \frac{4}{3}\pi r^3, \quad S = 4\pi r^2.$$

If $\frac{dV}{dt}$ is proportional to S , then $\frac{dr}{dt}$ becomes constant.

Integration

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1).$$

$$\int \frac{\log x}{x} dx = \frac{1}{2}(\log x)^2 + C.$$

Area between curves If upper curve is $y = f(x)$ and lower curve is $y = g(x)$,

$$A = \int_a^b [f(x) - g(x)] dx.$$

Vectors

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta.$$

Triangle inequality:

$$|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|.$$

Line in 3D

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} = t.$$

To check intersection of two lines, write both in parametric form and solve for the parameters.

Inequality with modulus

$$|X| \leq a \iff -a \leq X \leq a \quad (a \geq 0).$$

Area of triangle by determinant

$$\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}.$$

Linear programming Write variables, objective function, constraints, feasible region, corner points, and then evaluate the objective function at each corner point.